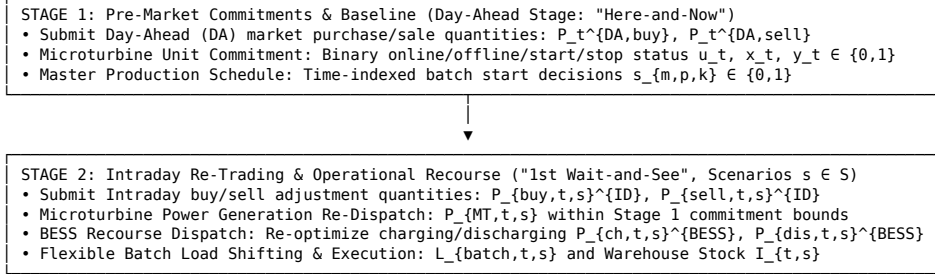


Multi-Stage Stochastic MILP Formulation for Industrial Factory, Microturbine, and BESS (Up to Intraday Market Stage)

1. Executive Overview & Multi-Stage Decision Architecture

This document presents the complete mathematical formulation for co-optimizing an **industrial batch manufacturing plant**, an on-site **Microturbine (MT)**, and a **Battery Energy Storage System (BESS)** across sequential electricity market stages. Building upon the baseline Day-Ahead model defined in `factory-mt-bess-dayahead-scheduling-v6md.pdf` [14], this formulation extends the architecture into a **multi-stage stochastic mixed-integer linear programming (MILP)** framework up to the **Intraday Market Stage**, strictly adhering to `industrial-large-consumer-multistage-optimizationmd.pdf` [15].

In accordance with market gate closures and progressive uncertainty realization, the decision timeline up to Intraday operations is structured into two main stages:



2. Sets, Parameters, and Decision Variables

2.1 Sets and Indices

- $t \in T = \{1, 2, \dots, T\}$: Hourly time periods over the scheduling horizon ($T = 24$ h).
- $s \in S$: Stochastic scenarios representing realizations of intraday prices and operational fluctuations.
- $m \in M$: Parallel industrial processing machines.
- $p \in P$: Production batch tasks to be executed.
- $k \in K$: Candidate batch start time slots $a_k = k\Delta$ (where $\Delta = 0.5$ h).
- $b \in B$: Piecewise power output blocks for the Microturbine.

2.2 Parameters

- π_s : Probability of scenario s ($\sum_{s \in S} \pi_s = 1$).
- $\lambda_t^{DA,buy}, \lambda_t^{DA,sell}$: Day-Ahead electricity purchase and sale prices (€/MWh).
- $\lambda_{buy,t,s}^{ID}, \lambda_{sell,t,s}^{ID}$: Intraday electricity adjustment purchase and sale prices in scenario s (€/MWh).
- $l_{base,t}$: Base factory non-shiftable electrical load (MW).
- $d_{m,p}$: Power demand of task p on machine m (MW).
- $td_{m,p}$: Processing duration of task p on machine m (hours).
- t_{buf} : Mandatory inter-batch setup/cooling buffer time on each machine (hours).
- Y_p : Product yield resulting from completing task p (units).
- δ_t : Hourly product delivery demand required from the warehouse (units/h).
- I_{max}, I_0 : Warehouse storage capacity and initial product stock (units).
- $P_{MT,min}, P_{MT,max}$: Minimum and maximum Microturbine electrical output (MW).
- w_b : Upper capacity bound for piecewise block b of the MT (MW).
- C_0 : MT no-load operating cost coefficient (€/h).

- C_b : Marginal fuel/operating cost for MT output block b (€/MWh).
- SUC, SDC : MT start-up cost and shut-down cost (€/event).
- RU, RD : MT ramp-up and ramp-down limits during continuous online operation (MW/h).
- SRU, SRD : MT start-up and shut-down ramping limits (MW/h).
- MUT, MDT : Minimum up-time and minimum down-time for MT (hours).
- $\bar{P}$: Maximum BESS charge/discharge power capacity (MW).
- $SoC_{min}, SoC_{max}, SoC_0$: Minimum, maximum, and initial State of Charge of the BESS (MWh).
- η_{ch}, η_{dis} : BESS charging and discharging efficiency factors ($0 < \eta < 1$).
- C_{TP} : BESS throughput degradation cost coefficient (€/MWh).
- Q_{buy}, Q_{sell} : Maximum Day-Ahead grid import/export capacity limits (MW).
- $Cap_{buy}^{ID}, Cap_{sell}^{ID}$: Maximum Intraday re-trading buy/sell capacity limits (MW).

2.3 Decision Variables

Stage 1: Pre-Market Commitments (“Here-and-Now”)

- $P_t^{DA,buy} \geq 0$: Day-Ahead power purchased from the grid in hour t (MW).
- $P_t^{DA,sell} \geq 0$: Day-Ahead power sold to the grid in hour t (MW).
- $u_t \in \{0, 1\}$: Binary status indicating if the Microturbine is online (1) or offline (0) in hour t .
- $x_t \in \{0, 1\}$: Binary start-up trigger for the Microturbine in hour t .
- $y_t \in \{0, 1\}$: Binary shut-down trigger for the Microturbine in hour t .
- $s_{m,p,k} \in \{0, 1\}$: Binary decision to start task p on machine m at time candidate a_k .

Stage 2: Intraday Operational Recourse (“1st Wait-and-See”, Scenario-Dependent)

- $P_{buy,t,s}^{ID} \geq 0$: Additional power purchased in the Intraday market in hour t , scenario s (MW).
- $P_{sell,t,s}^{ID} \geq 0$: Additional power sold in the Intraday market in hour t , scenario s (MW).
- $P_{MT,t,s} \geq 0$: Total active electrical generation from the MT in hour t , scenario s (MW).
- $P_{MT,b,t,s} \geq 0$: Power output generated in block b of the MT in hour t , scenario s (MW).
- $P_{ch,t,s}^{BESS} \geq 0$: BESS active charging power in hour t , scenario s (MW).
- $P_{dis,t,s}^{BESS} \geq 0$: BESS active discharging power in hour t , scenario s (MW).
- $SoC_{t,s} \geq 0$: BESS State of Charge at the end of hour t , scenario s (MWh).
- $v_{ch,t,s}, v_{dis,t,s} \in \{0, 1\}$: Binary BESS operational mode indicators (charging vs. discharging).
- $L_{batch,t,s} \geq 0$: Total factory industrial batch power consumption in hour t , scenario s (MW).
- $nprod_{t,s} \geq 0$: Quantity of finished product units completed in hour t , scenario s (units).
- $I_{t,s} \geq 0$: End-of-hour warehouse product inventory level in hour t , scenario s (units).

3. Multi-Stage Objective Function

The objective function minimizes total deterministic Stage 1 Day-Ahead commitment costs plus the expected Stage 2 Intraday trading adjustments, Microturbine operation and commitment costs, and BESS degradation expenses across all scenarios $s \in S$:

$$\min \sum_{t=1}^T \mathcal{C}_t^{DA} + \sum_{s \in S} \pi_s \sum_{t=1}^T (\mathcal{C}_{t,s}^{ID} + \mathcal{C}_{t,s}^{MT} + \mathcal{C}_{t,s}^{BESS})$$

Cost Component Formulations**1. Day-Ahead Energy Procurement Cost (€):**

$$c_t^{DA} = \lambda_t^{DA,buy} P_t^{DA,buy} - \lambda_t^{DA,sell} P_t^{DA,sell} \quad \forall t \in T$$

2. Intraday Re-Trading Adjustment Cost (€):

$$c_{t,s}^{ID} = \lambda_{buy,t,s}^{ID} P_{buy,t,s}^{ID} - \lambda_{sell,t,s}^{ID} P_{sell,t,s}^{ID} \quad \forall t \in T, \forall s \in S$$

3. Microturbine Fuel, Operating & Unit Commitment Cost (€):

$$c_{t,s}^{MT} = C_0 P_{MT,min} u_t + \sum_{b \in B} C_b P_{MT,b,t,s} + SUC \cdot x_t + SDC \cdot y_t \quad \forall t \in T, \forall s \in S$$

4. BESS Throughput Degradation Cost (€):

$$c_{t,s}^{BESS} = C_{TP} (P_{ch,t,s}^{BESS} + P_{dis,t,s}^{BESS}) \quad \forall t \in T, \forall s \in S$$

4. System Power Balance Constraint

For every hour $t \in T$ and scenario $s \in S$, total net power imported from the grid (Day-Ahead and Intraday) plus on-site generation and BESS discharge must exactly balance factory consumption (base load plus active batch load) and BESS charging:

$$(P_t^{DA,buy} - P_t^{DA,sell}) + (P_{buy,t,s}^{ID} - P_{sell,t,s}^{ID}) + P_{MT,t,s} + P_{dis,t,s}^{BESS} - P_{ch,t,s}^{BESS} = l_{base,t} + L_{batch,t,s} \quad \forall t \in T, \forall s \in S$$

5. Microturbine (MT) Constraints

Microturbine binary unit commitment decisions (u_t, x_t, y_t) are established in Stage 1 (“Here-and-Now”), while active power generation ($P_{MT,t,s}$) is adjusted dynamically in Stage 2 across scenarios.

5.1 Output Capacity & Piecewise Linearization

$$P_{MT,t,s} = P_{MT,min} u_t + \sum_{b \in B} P_{MT,b,t,s} \quad \forall t \in T, \forall s \in S$$

$$0 \leq P_{MT,b,t,s} \leq w_b u_t \quad \forall b \in B, \forall t \in T, \forall s \in S$$

5.2 Unit Commitment Logic & Status Consistency (Stage 1)

$$x_t - y_t = u_t - u_{t-1} \quad \forall t \in T \quad (u_0 = 0)$$

$$x_t + y_t \leq 1 \quad \forall t \in T$$

$$u_t, x_t, y_t \in \{0, 1\} \quad \forall t \in T$$

5.3 Minimum Up-Time and Minimum Down-Time Limits (Stage 1)

To ensure physical thermal stability, minimum up/down time windows are enforced across all hours $t \in T$ (truncated at $t = 1$):

$$\sum_{\tau=\max(1,t-MUT+1)}^t x_{\tau} \leq u_t \quad \forall t \in T$$
$$\sum_{\tau=\max(1,t-MDT+1)}^t y_{\tau} \leq 1 - u_t \quad \forall t \in T$$

5.4 Scenario-Dependent Inter-temporal Ramp Rate Limits (Stage 2)

Power output adjustments between consecutive hours are bounded by operational ramp rates and start-up/shut-down limits:

$$P_{MT,t,s} - P_{MT,t-1,s} \leq RU \cdot u_{t-1} + SRU \cdot x_t \quad \forall t \in T, \forall s \in S \quad (P_{MT,0,s} = 0)$$
$$P_{MT,t-1,s} - P_{MT,t,s} \leq RD \cdot u_t + SRD \cdot y_t \quad \forall t \in T, \forall s \in S$$

6. Battery Energy Storage System (BESS) Recourse Constraints

BESS charging, discharging, and State of Charge (SoC) profiles are re-optimized in Stage 2 to capture intraday market price spreads and compensate for batch load variations.

6.1 State of Charge Dynamics

$$SoC_{t,s} = SoC_{t-1,s} + \eta_{ch} P_{ch,t,s}^{BESS} \Delta t - \frac{1}{\eta_{dis}} P_{dis,t,s}^{BESS} \Delta t \quad \forall t \in T, \forall s \in S \quad (SoC_{0,s} = SoC_0)$$

6.2 Capacity Bounds & Terminal Energy Requirement

$$SoC_{min} \leq SoC_{t,s} \leq SoC_{max} \quad \forall t \in T, \forall s \in S$$
$$SoC_{T,s} \geq SoC_0 \quad \forall s \in S$$

6.3 Power Limits & Non-Simultaneous Operation

$$0 \leq P_{ch,t,s}^{BESS} \leq \bar{P} v_{ch,t,s} \quad \forall t \in T, \forall s \in S$$
$$0 \leq P_{dis,t,s}^{BESS} \leq \bar{P} v_{dis,t,s} \quad \forall t \in T, \forall s \in S$$
$$v_{ch,t,s} + v_{dis,t,s} \leq 1 \quad (v_{ch,t,s}, v_{dis,t,s} \in \{0, 1\}) \quad \forall t \in T, \forall s \in S$$

7. Industrial Batch Production & Warehouse Constraints

Adopting the precise **time-indexed batch scheduling formulation** from factory-mt-bess-dayah ead-scheduling-v6md.pdf [14]:

7.1 Single Task Assignment & Non-Overlap with Buffer

1. **Task Execution Bounds:** Each manufacturing task runs at most once across all available machines and start candidates:

$$\sum_{m \in M} \sum_{k \in K} s_{m,p,k} \leq 1 \quad \forall p \in P$$

2. **Machine Non-Overlap & Cooling Buffer:** A batch started at time a_k blocks machine m for all candidate start times $a_j \in [a_k, a_k + td_{m,p} + t_{buf})$:

$$\sum_{p \in P} \sum_{k: a_k \leq a_j < a_k + td_{m,p} + t_{buf}} s_{m,p,k} \leq 1 \quad \forall m \in M, \forall j \in K$$

7.2 Hourly Batch Power Load & Product Yield

Let $ov_{m,p,k,t}$ denote the exact continuous fraction of hour t occupied by task p started at candidate time $a_k = k\Delta$ on machine m :

$$ov_{m,p,k,t} = \max(0, \min(a_k + td_{m,p}, t) - \max(a_k, t - 1)) \in [0, 1]$$

The resulting active batch power load $L_{batch,t,s}$ and hourly product yield $nprod_{t,s}$ are:

$$L_{batch,t,s} = \sum_{m \in M} \sum_{p \in P} \sum_{k \in K} d_{m,p} \cdot ov_{m,p,k,t} \cdot s_{m,p,k} \quad \forall t \in T, \forall s \in S$$

$$nprod_{t,s} = \sum_{m \in M} \sum_{p \in P} \sum_{k \in K} \frac{Y_p}{td_{m,p}} \cdot ov_{m,p,k,t} \cdot s_{m,p,k} \quad \forall t \in T, \forall s \in S$$

7.3 Warehouse Product Inventory Balance

Finished product inventory accumulates from hourly manufacturing output and is drawn down by fixed hourly customer deliveries δ_t :

$$I_{t,s} = I_{t-1,s} + nprod_{t,s} - \delta_t \quad \forall t \in T, \forall s \in S \quad (I_{0,s} = I_0)$$

$$0 \leq I_{t,s} \leq I_{max} \quad \forall t \in T, \forall s \in S$$

$$I_{T,s} \geq I_0 \quad \forall s \in S$$

8. Trading Limits & Non-Anticipativity Constraints

8.1 Grid Trading Capacity Bounds

$$0 \leq P_t^{DA,buy} \leq Q_{buy}, \quad 0 \leq P_t^{DA,sell} \leq Q_{sell} \quad \forall t \in T$$

$$0 \leq P_{buy,t,s}^{ID} \leq Cap_{buy}^{ID}, \quad 0 \leq P_{sell,t,s}^{ID} \leq Cap_{sell}^{ID} \quad \forall t \in T, \forall s \in S$$

8.2 Day-Ahead Bidding Curve Monotonicity & Non-Anticipativity

To guarantee valid market bidding curves submitted prior to price realization: 1. **Monotonicity:**

$$P_t^{DA,buy}(\omega_1) \leq P_t^{DA,buy}(\omega_2) \quad \text{for scenarios where } \lambda_t^{DA}(\omega_1) \leq \lambda_t^{DA}(\omega_2)$$

2. **Non-Anticipativity:**

$$P_t^{DA,buy}(\omega_1) = P_t^{DA,buy}(\omega_2) \quad \text{for scenarios where } \lambda_t^{DA}(\omega_1) = \lambda_t^{DA}(\omega_2)$$

9. Mathematical Comparison & Variable Mapping

The table below summarizes how the baseline Day-Ahead model (factory-mt-bess-dayahead-scheduling-v6md.pdf [14]) is transformed into the multi-stage stochastic formulation up to Intraday (industrial-large-consumer-multistage-optimizationmd.pdf [15]):

Model Component	Baseline Day-Ahead Model (v6md)	Multi-Stage Intraday Model (multistage)	Decision Stage
Market Power Import/Export	P_t^{buy}, P_t^{sell}	$P_t^{DA,buy}, P_t^{DA,sell}$ (DA) / $P_{buy,t,s}^{ID}, P_{sell,t,s}^{ID}$ (ID)	Stage 1 (DA) / Stage 2 (ID)
MT Commitment Status	$u_t, x_t, y_t \in \{0, 1\}$	$u_t, x_t, y_t \in \{0, 1\}$ (Fixed across scenarios)	Stage 1 (Here-and-Now)
MT Output Power	$P_{MT,t}, P_{MT,b,t}$	$P_{MT,t,s}, P_{MT,b,t,s}$ (Re-dispatched per scenario)	Stage 2 (Operational Recourse)
BESS Charge/Discharge	$P_t^{ch}, P_t^{dis}, SoC_t$	$P_{ch,t,s}^{BESS}, P_{dis,t,s}^{BESS}, SoC_{t,s}$ (Recourse dispatch)	Stage 2 (Operational Recourse)
Batch Schedule Decisions	$s_{m,p,k} \in \{0, 1\}$	$s_{m,p,k} \in \{0, 1\}$ (Fixed baseline master schedule)	Stage 1 (Here-and-Now)
Factory Batch Load	$L_{batch,t}$	$L_{batch,t,s}$	Stage 2 (Operational Recourse)
Warehouse Inventory	I_t	$I_{t,s}$	Stage 2 (Operational Recourse)

References

1. **factory-mt-bess-dayahead-scheduling-v6md.pdf** [14]: Mixed-Integer Linear Programming (MILP) model for co-optimizing industrial batch scheduling, Microturbine, and BESS in the Day-Ahead market.
2. **industrial-large-consumer-multistage-optimizationmd.pdf** [15]: Multi-stage stochastic optimization framework for large industrial consumers in Day-Ahead, Intraday, and Balancing markets.